

Focused Hypothesis Report — Horse PTPRN2 (A0A9L0T4W6) and "protein dephosphorylation" (GO:0006470)

Hypothesis evaluated: The horse protein A0A9L0T4W6 participates in protein dephosphorylation (GO:0006470). **Focus type:** function_assignment. **Target:** *Equus caballus* (NCBITaxon:9796), UniProt A0A9L0T4W6, gene label PTPRN2. **Date:** 2026-09-08.

Executive Judgment

Verdict: REFUTED / over-annotated for a direct function assignment.

Provenance verified (Iteration 2): The supplied 976-aa sequence's SHA-256 was recomputed in-sandbox = 6e469207454cac638b007421577b3c8140a28f2f24ed1e0ee900486ad5d7c821 — **exact match** to the frozen spec. The live UniProt record A0A9L0T4W6 (unreviewed TrEMBL, name "Protein-tyrosine-phosphatase") carries **only one MF GO term, GO:0004725 (protein tyrosine phosphatase activity), with evidence code IEA:InterPro** — i.e., an automated domain-presence annotation, no experimental support, and GO:0006470 is a *downstream* BP inference from it. InterPro correctly types the protein as IPR033522 "IA-2/IA-2_beta" (pseudophosphatase family).

Horse A0A9L0T4W6 is the ortholog of human **PTPRN2 (IA-2 β / phogrin)**, a member of the receptor-type protein-tyrosine-phosphatase (RPTP) family that is one of the best-characterized **pseudophosphatases**. Across multiple independent biochemical studies the wild-type IA-2 family PTP-like domain has **no activity against conventional (protein/phosphotyrosine) substrates**; catalytic competence can only be engineered by "back-mutating" three non-consensus active-site residues to the active PTP1B identities (Drake 2003,).

P 12697028

The **only demonstrated intrinsic catalytic activity** of phogrin/PTPRN2 is toward

phosphoinositide lipids (PI3P, PI(4,5)P2 — Caromile 2010,
 ;
 P 20097759
 Sengelaub 2016,
),
 P 26620550
 which is **lipid dephosphorylation, a process distinct from GO:0006470**. The protein's
 documented protein-directed effects are **non-catalytic** — scaffolding of the insulin receptor/
 IRS2 (Torii 2018,
)
 P 29483197
 and heterodimeric **down-regulation** of an active RPTP (Gross 2002,
),
 P 12364328
 i.e., *negative regulation* of dephosphorylation, not participation in it.

Most important caveats. (1) All functional data are from human/rodent/*C. elegans* orthologs; there is **no horse-specific assay**. Transfer is justified because the horse sequence is unambiguously the PTPRN2 ortholog and conserves the catalytic-degeneracy signature (see Identity below). (2) "Lack of intrinsic catalysis does not exclude participation" was tested explicitly — the only regulatory link that is directional *opposes* dephosphorylation, so it does not rescue the seed hypothesis.

Evidence Matrix

#	Citation (PMID/ DOI)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
1	Drake 2003 (12697028)	Direct enzyme assay + mutagenesis	Refutes direct role	Does IA-2/IA-2 β dephosphorylate proteins?	WT IA-2 & IA-2 β "lack activity against conventional PTP substrates"; activity needs back-mutation of 3 residues (PTP1B Tyr46/Asp181/Ala217)	Recombinant human catalytic domains, in vitro
2	Caromile 2010 (20097759)	Direct enzyme assay (lipid)	Qualifies / Refutes (protein)	What is phogrin's substrate?	"no significant enzymatic activity for phogrin has ever been reported"; phogrin dephosphorylates PI3P and PI(4,5)P2, not PI(3,4,5)P3; Cys- dependent	Recombinant overexpression β -cell biology
3	Sengelaub 2016 (26620550)	Functional/ enzymatic (cells)	Qualifies (lipid, not protein)	Substrate of PTPRN2 in cells	PTPRN2 reduces plasma-membrane PI(4,5)P2, releasing cofilin $\rightarrow$ actin remodeling/ metastasis	Metastatic human breast cancer cells
4	Gross 2002 (12364328)	Interaction (Y2H/pulldown/ co-IP)	Refutes/ competing	Does the PTP- like domain catalyze? Regulatory link?	"single enzymatically inactive PTP-like domain"; heterodimerizes with RPTP α/ϵ and down- regulates RPTP α activity	COS-1, yeast
5	Torii 2018 (29483197)	Mutant phenotype / interaction	Qualifies (non- catalytic)	Mechanism in signaling	"pseudophosphatase"; phogrin-IR complex stabilizes IRS2; KO lowers IRS2 44%	Mouse islets, β - cell lines

#	Citation (PMID/ DOI)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
6	Cai 2009 (19343207)	Genetic/ phenotype	Supports "inactive"	Nature of IA-2	IA-2 = "a catalytically inactive protein phosphatase" in dense-core-vesicle membranes	<i>C. elegans</i> IDA ortholog
7	This report — sequence analysis	Structural/ evolutionary (computational)	Supports transfer + refutes	Is horse the PTPRN2 ortholog? Is the catalytic site degenerate?	PTP-domain identity 80.7% to PTPRN2 vs 67.1% to PTPRN; P-loop and catalytic motifs identical to human PTPRN2; Asp sits at the PTP1B-Ala217 position (one of Drake's 3 non-consensus residues)	In-sandbox, UniProt Q92932/Q1684
8	This report — UniProt record	Database provenance	Refutes / context	Origin of the phosphatase annotation	Only MF term GO:0004725 with evidence IEA:InterPro ; no experimental GO; InterPro types it IPR033522 IA-2/ IA-2_beta; SHA-256 of sequence matches frozen spec exactly	Live UniProt A0A9L0T4W6 (2026-09-08)
9	This report — PTP1B alignment	Structural/ evolutionary (computational)	Refutes	Is the catalytic machinery intact in the horse itself?	Anchored to active PTP1B (P18031): P-loop HCSDGAGR vs active HCSAGIGR (Asp at Ala217 position); general-acid loop WYDQGVP vs active WPDFGVP (Pro→Tyr,	In-sandbox, UniProt A0A9L0T4W6 vs P18031

#	Citation (PMID/ DOI)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
					WPD→WYD); catalytic Cys897/invariant Arg retained. The isolated WPD at idx 811 is 82 aa from the P-loop (canonical spacing ~33) and is NOT the catalytic loop	

GO Term Scope Verification (QuickGO, retrieved Iteration 3)

Authoritative GO definitions confirm that the seed term is **protein-substrate-specific** and distinct from the lipid term that phogrin's activity actually fits:

- **GO:0006470** protein dephosphorylation (BP): *"The process of removing one or more phosphoric residues from a protein."*
- **GO:0004725** protein tyrosine phosphatase activity (MF): *"Catalysis of the reaction: protein tyrosine phosphate + H₂O = protein tyrosine + phosphate."*
- **GO:0046856** phosphatidylinositol dephosphorylation (BP): *"...removing one or more phosphate groups from a phosphatidylinositol."*
- **GO:0052866** phosphatidylinositol phosphate phosphatase activity (MF).

Because the only demonstrated PTPRN2 catalytic activity is on **phosphatidylinositol lipids**, it satisfies GO:0046856/GO:0052866 — **not** GO:0006470/GO:0004725, whose definitions explicitly require a protein substrate. This closes the decisive question's phosphoprotein-vs-phosphoinositide substrate separation.

GO Decision Table (leads — require curator verification)

GO term	Aspect	Current basis	Evidence verdict	Recommended action
GO:0006470 protein dephosphorylation	BP	Downstream of the MF PTP-activity call	Refuted (pseudophosphatase; no protein-substrate activity)	Do not assign / remove
GO:0004725 protein tyrosine phosphatase activity	MF	IEA: InterPro (domain presence)	Refuted for horse ortholog (WT inactive; ≥3 back-mutations needed)	Remove or down-qualify; not experimentally supported
GO:0052866 / GO:0004438 phosphatidylinositol/PI-phosphatase activity	MF	Not in record	Supported in human/mouse ortholog (Caromile 2010; Sengelaub 2016)	Candidate ISS/ISO add with low-activity caveat
GO:0046856 phosphatidylinositol dephosphorylation	BP	Not in record	Supported in ortholog	Candidate ISS/ISO add (replaces protein-dephos claim)
GO:0030141 secretory granule / GO:0030658 transport vesicle membrane	CC	IEA	Consistent with DCV biology	Retain (secondary to this hypothesis)

GO Curation Implications

- **GO:0006470 (protein dephosphorylation, BP):** Evidence does **not** support annotating horse PTPRN2 to this term as a direct/positive function. Recommended curator action: **do not assert / remove or restrict** any GO:0006470 (and the parent MF GO:0004721 phosphoprotein phosphatase activity / GO:0004725 protein tyrosine phosphatase activity)

annotation that is based only on family membership or automated pipelines (e.g., InterPro/ARBA "PTP domain → protein dephosphorylation"). This is classic **pseudophosphatase over-annotation by paralog/domain inheritance**.

- **More accurate MF terms (leads, if any activity is to be annotated):** phosphatidylinositol phosphatase / phosphoinositide phosphatase activity (e.g., **GO:0004438** / **GO:0052866**) with **BP GO:0046856 phosphatidylinositol dephosphorylation** — supported by Caromile 2010 and Sengelaub 2016, but note the activity is weak/context-dependent and should carry an appropriate evidence code (IDA in human/mouse; ISS/ISO for horse) and possibly a "low activity" note.
- **CC:** insulin/dense-core secretory-granule membrane and plasma membrane localization are well supported (secondary to this hypothesis).
- **Non-core / regulatory:** any protein-phosphorylation link is best captured as **negative regulation of a PTP** or **scaffolding (protein binding)** — but avoid "protein binding" as a terminal recommendation; prefer the phosphoinositide MF/BP terms where activity is annotated.

Bottom line for the curator: the seed function assignment (protein dephosphorylation) should be treated as **incorrect/over-precise**; the defensible catalytic annotation, if any, is lipid (phosphoinositide) dephosphorylation.

Mechanistic Scope

Immediate molecular function tested = enzymatic removal of phosphate from a **protein** amino-acid residue. Direct gene-product activity: **absent** for proteins (degenerate PTP active site). The gene product's genuine molecular activities/roles are: (i) **weak phosphoinositide phosphatase** (lipid head-group, not protein); (ii) **secretory-granule membrane scaffold** influencing insulin/IRS2 signaling; (iii) **autoantigen** in diabetes. Downstream phenotypes (insulin secretion, β -cell signaling, cancer-cell migration) are **consequences of lipid signaling and scaffolding**, not evidence of protein-dephosphorylation catalysis.

Conflicts and Alternatives

- **Paralog confusion:** IA-2 (PTPRN, Q16849) vs IA-2 β /phogrin (PTPRN2, Q92932). Sequence analysis assigns the horse protein to **PTPRN2** (80.7% vs 67.1% PTP-domain identity; identical P-loop/WPD-loop to PTPRN2). Both paralogs are nonetheless pseudophosphatases, so the conclusion is robust to residual paralog uncertainty.
- **"Domain present $\Rightarrow$ function present" fallacy:** the horse GO:0004725 call is IEA:InterPro, driven purely by PTP-domain detection; the InterPro rule even asserts an active-site phosphocysteine at residue 897. Yet the catalytic Cys/invariant Arg are embedded in a **degenerate active site** (P-loop **HCS DGAGR**; general-acid loop **WYD QGV P** not **WP DFGVP**), and direct assays (Drake 2003; Caromile 2010) show the WT domain is catalytically dead on proteins. Retention of a Cys/Arg (and a coincidental non-catalytic **WPD** triplet 82 aa away) is not evidence of activity.
- **Substrate misassignment:** literature activity is on **phosphoinositides**, which can be mis-mapped to "dephosphorylation $\rightarrow$ protein dephosphorylation" if the lipid nature of the substrate is not tracked.
- **Directionality:** the one directional protein-phosphorylation link (Gross 2002) is **inhibitory** toward an active PTP — opposite to "performing protein dephosphorylation."

Knowledge Gaps

1. **No horse-specific biochemical data.** Checked: PubMed. Matters because the annotation is for *E. caballus*. Resolve: express horse PTP-like domain and assay pNPP/phosphopeptide vs PI3P/PI(4,5)P2.
2. **Exact active-site mapping in horse. Partially resolved (Iteration 2):** anchored alignment to active PTP1B (P18031) confirms a degenerate P-loop (**HCS DGAGR**, Asp at the PTP1B-Ala217 position) and an altered general-acid loop (**WYD QGV P** vs **WP DFGVP**; Pro $\rightarrow$ Tyr, WPD $\rightarrow$ WYD), with the catalytic Cys897/invariant Arg retained. The isolated **WPD** triplet noted in Iteration 1 is 82 aa from the P-loop (outside the $\sim$ 33-aa catalytic register) and is not the catalytic loop. Residual gap: a full gapped alignment/AlphaFold superposition would fix each Drake residue (Tyr46/Asp181/Ala217 equivalents) unambiguously.
3. **Physiological relevance and magnitude of lipid-phosphatase activity in vivo** remains debated (low specific activity in some assays). Resolve: catalytic-Cys mutant rescue experiments in horse-relevant tissue.

Discriminating Tests

1. **In vitro dual-substrate assay** of recombinant horse PTP-like domain: phosphotyrosine peptide/pNPP (protein-like) vs PI3P/PI(4,5)P2 (lipid). Prediction: negligible protein activity, measurable lipid activity → refutes GO:0006470, supports lipid term.
2. **Active-site back-mutation** (horse equivalents of PTP1B Tyr46/Asp181/Ala217): if only the triple mutant gains protein-pTyr activity, WT is confirmed pseudophosphatase.
3. **Catalytic-Cys (C→S) mutant** in a cell PI(4,5)P2 reporter assay to confirm lipid- (not protein-) directed catalysis.
4. **AlphaFold/structure superposition** on PTP1B to verify WPD-loop general-acid geometry and the Ala217-position Asp.

Curation Leads (require curator verification)

- **Action change:** Flag/remove GO:0006470 (and MF GO:0004725/GO:0004721) as a direct annotation for A0A9L0T4W6 if present via automated/ISS routes; treat as **pseudophosphatase over-annotation**.
- **Candidate replacement terms (leads):** MF phosphoinositide/PI phosphatase activity (GO:0052866 / GO:0004438); BP GO:0046856 (phosphatidylinositol dephosphorylation) — annotate by ISS/ISO from human/mouse with a low-activity caveat, if the curator wishes to record catalytic function at all.
- **Candidate references + verify snippets:**
- Drake 2003,

P 12697028

— "characterized by a lack of activity against conventional PTP substrates"; three-residue back-mutation restores activity.

- Caromile 2010,

P 20097759

— "no significant enzymatic activity for phogrin has ever been reported"; dephosphorylates PI3P and PI(4,5)P2.

- Gross 2002,

P 12364328

— "single enzymatically inactive PTP-like domain"; "down-regulate RPTPalpha enzymatic activity".

- Sengelaub 2016,

P 26620550

— PTPRN2 reduces plasma-membrane PI(4,5)P2.

- Torii 2018,

P 29483197

— "pseudophosphatase"; phogrin-IR/IRS2 scaffolding.

- Cai 2009,

P 19343207

— IA-2 "a catalytically inactive protein phosphatase".

- **Suggested curator questions:** Is the existing GO:0006470 annotation IEA/ISS-derived from PTP-domain presence? Does any experimental annotation for the horse entry exist? Should catalytic annotation be recorded as lipid rather than protein dephosphorylation?
- **Suggested experiments:** the dual-substrate and back-mutation assays above.

Limitations

- No gapped global aligner or structure tool in the execution sandbox; identity computed from an anchored ungapped PTP-domain window and whole-protein k-mer Jaccard. Paralog assignment is nonetheless clear.
- All primary functional evidence is from non-horse orthologs; horse conclusions are **justified transfer**, explicitly labeled, not direct observation.
- SHA-256 of the supplied sequence was recomputed with a pure-Python implementation (hashlib/struct blocked) and **matches the frozen spec exactly** (`6e46920745...5d7c821`; length 976), so the analysis is on the intended sequence.

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